

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	B. Tech AI&DS 2025	Date:	23.07.26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1	40
		Q2,Q3	60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS**Q1. Convex Hull – Outer Shape of Surveyed Coordinates**

Apply the Brute Force Convex Hull algorithm on the set $\{(0,1), (2,6), (5,8), (8,6), (9,2), (6,0), (3,1), (4,4)\}$. Clearly interpret the problem and explain the purpose of finding the convex hull. Check all possible line pairs and determine which edges form the external boundary. Show the logical steps used to separate hull points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in land survey analysis.

Q2. Convex Hull – Brute Force Approach

Apply the Brute Force Convex Hull algorithm on the set $\{(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)\}$. Interpret the problem and explain the concept of convex hull. Check all possible line pairs and identify valid hull edges. Show the logical steps used to determine the boundary. Count the total comparisons performed. Analyze the time complexity of the method. Interpret the result and its practical significance.

Q3. Linear Search – Search for Middle-Late Position Target

Given the unsorted dataset [22, 31, 47, 58, 16, 39, 64], apply Linear Search to find the element 39. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears in the latter half of the dataset.

2/3

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Linear Search - Repeated Value Search :

Problem Interpretation :

Given the sorted dataset :

[10, 20, 30, 40, 50, 60, 70]

Find the element 35 using Binary search.

Since the dataset is already sorted, Binary search is chosen because it repeatedly divides the search interval into halves.

Algorithm used :

Binary search repeatedly compares the target with the middle element.

→ If target < mid $\Rightarrow$ search left half.

→ If target > mid $\Rightarrow$ search right half.

→ If continue until search interval empty.

Step by Step Search Process :

Step 1: Low = 0, high = 6, mid = $\frac{0+6}{2} = 3$

mid = 40

Comparison: $35 < 40 \rightarrow$ search left half.

New range:

Low = 0

high = 2

Step 2: $Low = 0$, $High = 2$, $mid = 2$

Comparison : $35 > 20$

Search Right half

New range: $Low = 2$, $high = 2$

Step 3: $Low = 2$, $High = 2$, $mid = 2$

Comparison : $35 > 30$, Search right half.

Then $Low > High$ Search stops.

* No. of comparisons:

Comparison with 40

Comparison with 20

Comparison with 30

Total Comparison = 3.

* Result:

Target element = 35

Status = Element not found.

* Time complexity:

• Best case : $O(1)$

• Average case : $O(n)$

• Worst case : $O(n)$

* Verification:

the first occurrence of 26 is correctly found at index 1.

Binary Search - Unsuccessful Search:
Dataset : [10, 20, 30, 40, 50, 60, 70]
target : 35.

* Algorithm used :

- If target < mid $\rightarrow$ search left half
- If target > mid $\rightarrow$ search right half

* Step by step search process :

Step 1 : Low = 0

High = 6

mid = 3

Comparison : $35 < 40$

search left half

New range : Low = 0 High = 2

Step 2 : Low = 0

High = 2

mid = 1

comparison : $35 > 20$

search right half

New range : Low = 2 High = 2.

Step 3 : Low = 2

High = 2

mid = 2

comparison : $35 > 30$

search right half.